

Windows & Linux Artifact Forensics

The "Artifact Sandwich" Mindset

Volatile = lost at reboot; but PERSISTENT artifacts survive FOREVER and tell you a LOT even without memory:

Windows Artifacts Cheat-Sheet

Artifact	Path / Source	What It Proves
Prefetch	\Windows\Prefetch*.pf	programs actually RUN (last executed times)
Event Logs	Security/System/Application	logons (4624), failures, service installs
Registry	NTUSER.DAT, SAM, SYSTEM	autostart (Run keys!), recent docs, MRU, uninstall list
AmCache	\Windows\AppCompat	executable install/execution history
Shellbags	Registry	folders the user opened (the 'visited folders')
LNK files	recent items	shortcuts opened
Scheduled Tasks	\Windows\Tasks	persistence + timing
BITS/jobs	BITS queue	unknown-file downloads
\$MFT / \$J	NTFS	file creation/deletion/residency

Reading the Autostart (Persistence Hunt)

```
HKCU\Software\Microsoft\Windows\CurrentVersion\Run
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
```

HKLM\...\Services # service persistence

An entry pointing at \Users\Public\update.exe = persistence found.

Linux Artifacts

Artifact	Location	Value
bash_history	~/.bash_history	commands the user ran
auth.log/syslog	/var/log	logons, su/sudo, cron
cron	/etc/cron*, spool	persistence
last/logins	/var/log/wtmp	logon history
.bashrc/.profile	home	sneaky auto-start
/tmp	...	dropped tools

```
grep -E "sudo|su " ~/.bash_history
last -x
find /etc/cron* -type f -mtime -30
```

The Wall of Rebuilt Facts

Artifacts let you answer: "What exactly was done, by whom, on this box?" – without memory.

Prefetch says "powershell ran at 03:14"

Event log says "logon user 'bob', source IP 10.0.0.9, 03:13"

Run key says "and it comes back at boot"

Together = the timeline (next lesson).

Tooling

Autopsy / Sleuth Kit – disk + artifact analysis GUI

Log2timeline / plaso – build super-timelines (everything, sorted)

KAPE (toolkit) – collect artifacts fast at scale

PowerShell/reg.exe – quick triage on a live Windows box

The analyst's question: every file has a story. Artifacts are the breadcrumbs arranged by the OS itself. Know where they live and you can always rebuild the story.

[← Memory Forensics with Volati...](#)

[Timeline Analysis & Forensic R...](#)